EWP situations — review sheet

You are on site and see each of the following situations. State whether or not it is allowed, and your specific reasoning.

	Situation	Acceptable yes/no	Document	Page	Detail/Chart	Correct Answer
1	Drawings call for 9.5" TJI S31. The flanges on site are 2.25" wide.					
2	Drawings call for 11.875" TJI S31 @ 19.2" OC, 3/4" subfloor, 40+25psf, directly applied ceiling, continuous span = 18'-5". Client has decided to install a dropped ceiling rather than gypsum ceiling.					
3	Drawings call for 16" TJI S31 roof joists @ 24"OC, 8/12 pitch. Contractor could only get 18".					
4	Drawings call for 3.5" x 3.5" Parallam 1.8E columns. Contractor substituted 3.5" x 3.5" Microllam 2.0E.					
5	Drawings call for 3.5" x 5.5" Timberstrand 1.3E lintel in plank orientation for interior sliding door. Carpenters drilled 1/4" holes in lintel to install the sliding door hardware.					
6	Foundation is 36' wide. Drawings call for 11.875" TJI S31 @ 19.2" OC, 3/4" subfloor, 40+25psf, directly applied ceiling, continuous span = 18'-0". The longest joists available from the supplier are 20'.					
7	Drawings call for 14" TJI S37. The flanges on site are 2.5" wide.					
8	Drawings call for a 0.5/12 slope for a renovation using 14" roof joists. Contractor ordered 16" joists, so now the slope is 0.125/12.					
9	Drawings call for 8' high 3.5" x 3.5" Timberstrand 1.3E column on a wood plate. Contractor installed a base plate with the screws perpendicular to the wide face of the strands.					
10	Drawings call for 3.5" x 11.875" Timberstrand 1.55E beam. The beam is 24' long with a column in the middle. Plumbers drilled a 3" diameter hole in the middle of the beam depth, directly above the centre of the interior column, then drilled a 2" hole 5" to the left.					
11	Drawings call for 11.875" TJI S31 @ 16" OC, 5/8" subfloor, 40+15psf, directly applied ceiling, simple span = 16'-11". Client has decided to install a dropped ceiling rather than gypsum ceiling.					

	Situation	Acceptable yes/no	Document	Page	Detail/Chart	Correct Answer
12	Drawings call for 16" TJI S31 roof joists @ 24" OC, 7/12 pitch, 25LL + 15DL, spanning 23'-9".					
13	Drawings call for 5.25" x 14" Microllam 2.0E instead of a girder truss to support the valley framing. The beam sits on a 2×6 wall. The factored reaction is 10,500#. The tapered end cut is 5.5" to stay under the sheathing.					
14	Drawings call for 2 ply 1.75" x 11.875" Microllam 2.0E with a 12' clear span. Plumber drilled a 2" hole at 5', 7' and 9' in the middle of the beam depth.					
15	Drawings call for 9.5" TJI S31 @ 19.2" OC, 5/8" subfloor, 40+15psf, no directly applied ceiling, simple span = 14'-6". Client asks you if they will be happy with the floor performance.					
16	Drawings call for 16" TJI S31 roof joists @ 24" OC, 5/12 pitch, 25LL + 15DL, spanning 23'-9". Carpenters bevelled the lower wall plate, installed shear blocking and used variable slope hangers and strap ties at the ridge beam.					
17	Drawings call for 5.25" x 14" Microllam 2.0E. The beam is side-loaded with a factored load of 400plf. The carpenter installed 3 rows of 0.128" x 3" nails @ 12" OC in the outside plies.					
18	Drawings call for 14' long 11.875" TJI S31 @ 16" OC. Electrician drilled a series of 1" diameter holes in the middle of the web depth from 4' to 10' @ 6" OC.					
19	Foundation is 32' wide. Bearing wall is at 22'. Drawings call for 14" TJI S31 @ 16" OC, 5/8" subfloor, 40+15psf, directly applied ceiling, continuous span.					
20	Drawings call for 14" S47 joists. Carpenters cut squash blocks 14 1/16" long.					
21	Carpenter cut 6" seat for a bird's mouth for a 2×6 wall. Seat cut extends 1/2" past inside face of wall.					
22	Drawings call for 3.5" x 11.875" Timberstrand 1.55E beam. The beam is 24' long with a column in the middle. The end bearing requirement is 1.9". The cap on the interior column is 6" long.					

	Situation	Acceptable yes/no	Document	Page	Detail/Chart	Correct Answer
23	Drawings call for 14' long 11.875" TJI S31 @ 16" OC. Joists are cantilevered 5" over brick, requiring an E6 detail. Electrician drilled 1/2" diameter holes in the cantilevers to run cable for exterior security cameras.					
24	Drawings call for 14" S47 joists. Carpenters cut web stiffeners 13 7/8" long.					
25	Drawings call for 9.5" TJI S31. Joists are cantilevered 4" for brick and support the roof trusses. Backspan is only 24".					